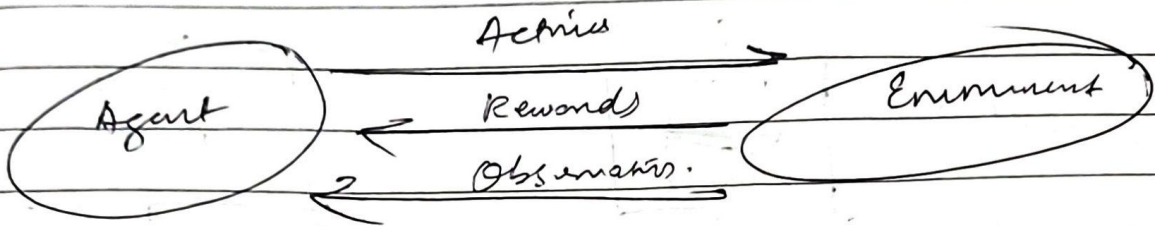
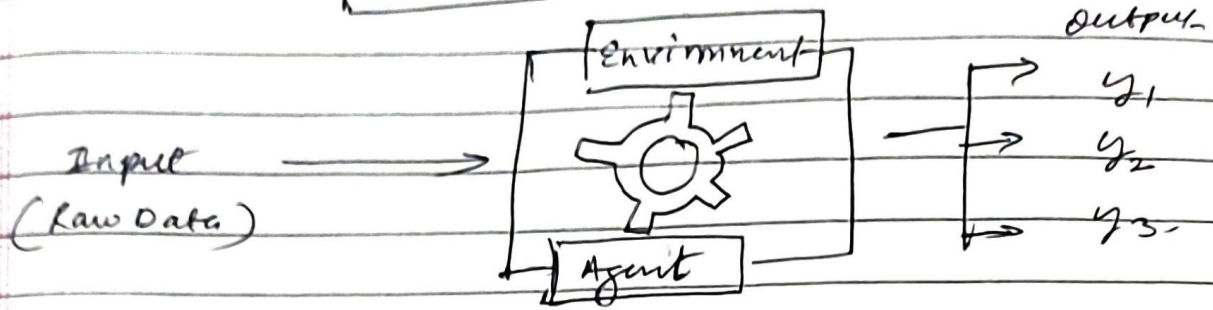




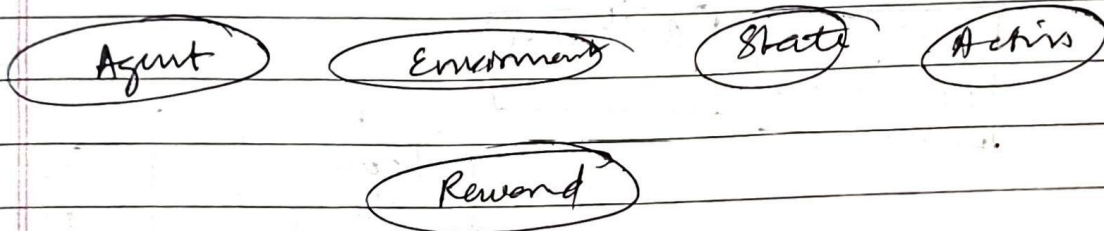
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REINFORCEMENT & REPRESENTATION LEARNING

Module-1



4 Main Elements of RL:-



+ve reward $\rightarrow$ Good Action
-ve reward $\rightarrow$ Bad Action

winning a chess piece = +10
losing a chess piece = -10
Win the game = +100.

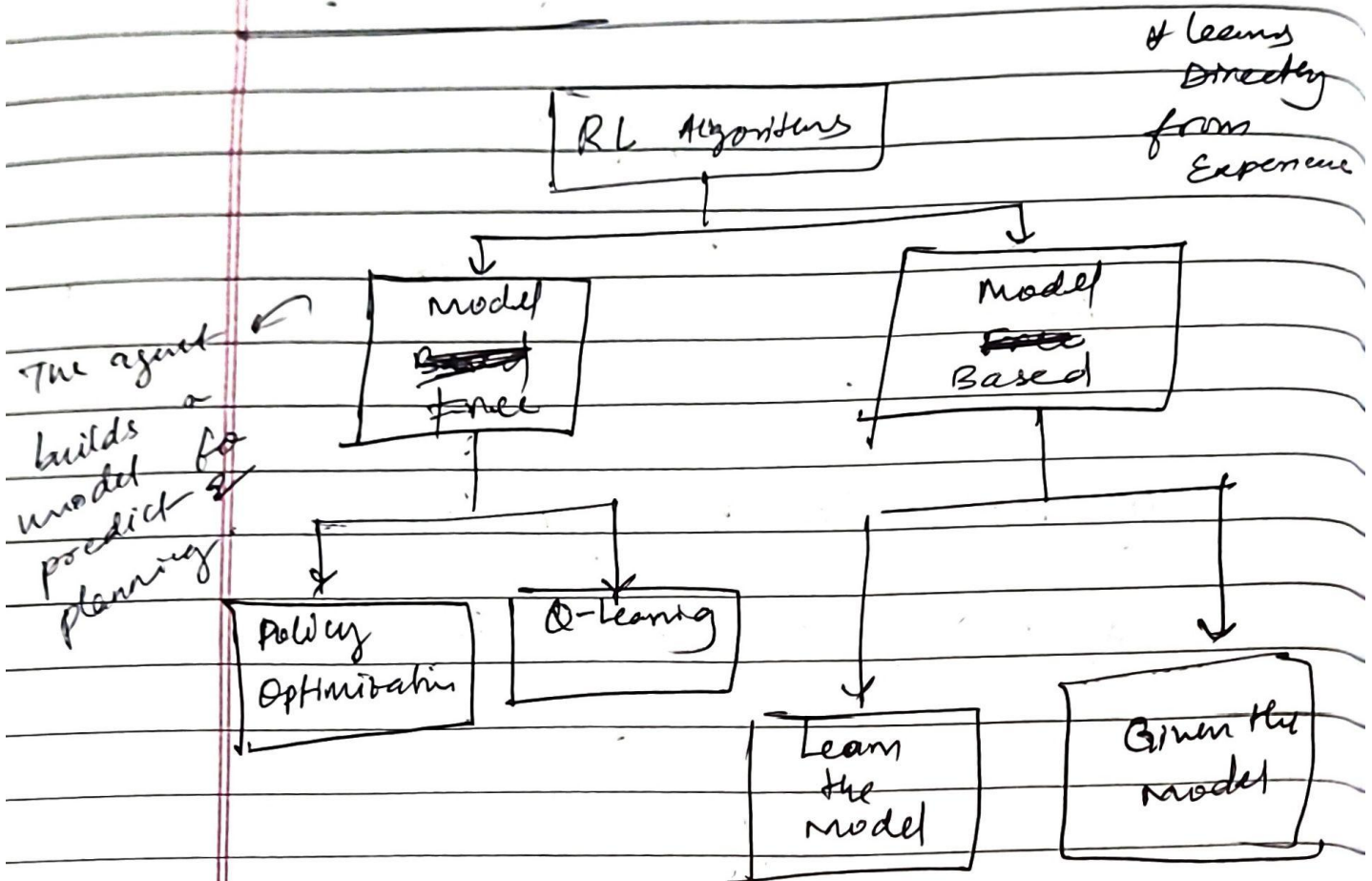
(*)
Policy: (like a rulebook)

- * It's the agent's strategy
- * Tells agent which action to take in a particular state.

Value Function (V)

Estimates how good a state is. It predicts total future reward from that state.

Q-Value : (or Action-Value Function).
Estimate how good a particular action is in a given state.



→ Markov Decision Process:

Defined by tuple:

$\langle S, A, R, P, \gamma \rangle$

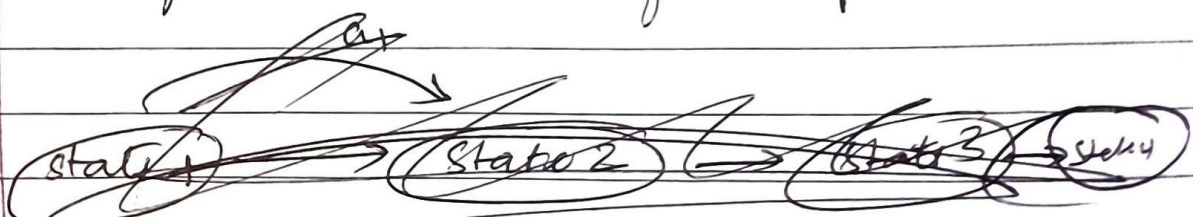
$S \rightarrow$ set of states

$A \rightarrow$ set of actions

$R \rightarrow$ Reward function

$\gamma \rightarrow$ Discount factor

$P \rightarrow$ State transition probabilities



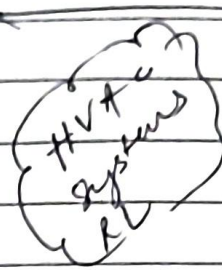
→ Popular RL Algorithms :

Q-Learning

SARSA (State-Action-Reward-State-Action)

Deep Q-Networks (DQN).

Policy Gradient Methods.

Represent Representation Learning	Extracts meaningful patterns from raw data	
Reinforcement Learning		
★ Its about understanding data.	Its about learning what action to take.	
★ It figures out what parts of the data matter.	★ It finds best actions to take in a given situation.	

Module - 2

→ Markov Decision Process (MDP)

"The future depends on the current state not the past state" → principle of MDP.

★ Main Components :

→ Actors & Critics :

Leads policies to select the Best Action

Leads value function (or Q value) that estimate value of current policy

→ Bellman's Equation: (To find value of a state)

* It says if there is a node nearest to the goal node, then assign that node with max rewards. Similarly reduce the rewards of the other nodes based on the proximity.

Value of a state	=	Reward of current state	+	Expected Value of Next state
------------------	---	-------------------------	---	------------------------------

Mathematically, This does not mean summation only

$$V(s) = \sum [P_{ts} (R_t + \gamma V(s'))]$$

Probability

or

$$V(s) = \max_a [R(s,a) + \gamma V(s')]$$

Reward at particular state & Action

Discount factor

value of next state

⇒ Bellman Q Value Formula:

$$Q(s,a) = R + \gamma \max_{a'} (Q(s',a'))$$

TRAFFIC SIGNAL CONTROLLER

States	Action	$\gamma = 0.1$
s_1 : Normal	a_1 (Keep Green)	
s_2 : Medium	a_2 (Switch signal)	
s_3 : Heavy		

Current state : (s_2)

Action a_1 :	Action a_2 :
to s_1 : $P=0.4$ $R=6$	to s_1 : $P=0.6$ $R=7$
to s_2 : $P=0.4$ $R=4$	to s_2 : $P=0.3$ $R=3$
to s_3 : $P=0.2$ $R=1$	to s_3 : $P=0.1$ $R=2$

Given Current value estimates :

$$V(s_1)=10, V(s_2)=6, V(s_3)=2.$$

~~$$Q(s,a) = \sum p(R + \gamma V(s'))$$~~

$$Q(s,a) = \sum \left[E \left[R_t + \gamma V(s) \right] \right]$$

2) Finding for action a_1 : $Q(s_2, a_1)$

~~$$Q(s_2, a_1) = \sum p \left[R_t + \gamma V(s') \right]$$~~

$$s_1 = 0.4 \left[6 + 0.1(10) \right] = 2.8$$

$$s_2 = 0.4 \left[4 + 0.1(6) \right] = 1.84$$

$$s_3 = 0.2 \left[1 + 0.1(2) \right]$$

$$Q(s_2, a_1) = 2.8 + 1.84 + 0.24 = 4.84$$

$$\hat{Q}(s_2, a_1) = 4.88$$

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Similarly, Now for $\hat{Q}(s_2, a_2)$.

$$\hat{Q}(s_2, a_2) = E[R + \gamma V(s_3)]$$

$$\begin{aligned} s_1 &: 0.6 [7 + 0.1(10)] \\ s_2 &: 0.3 [3 + 0.1(6)] \\ s_3 &: 0.1 [2 + 0.1(2)] \end{aligned}$$

$$s_1 = 4.8$$

$$s_2 = 1.08$$

$$s_3 = 0.22$$

$$\hat{Q}(s_2, a_2) = 4.8 + 1.08 + 0.22$$

$$\hat{Q}(s_2, a_2) = 6.1$$

Now, we get that:

$$\hat{Q}(s_2, a_1) < \hat{Q}(s_2, a_2)$$

$$4.88 < 6.1$$

Max. Q value = max rewards
meaning.

∴ Agent will select a_2

✓✓

⇒ SARSA : (State-Action-Reward-Action-State)
(On-policy).

$$Q(S, A) \leftarrow Q(S, A) + \alpha [R + \gamma Q(S', A') - Q(S, A)]$$

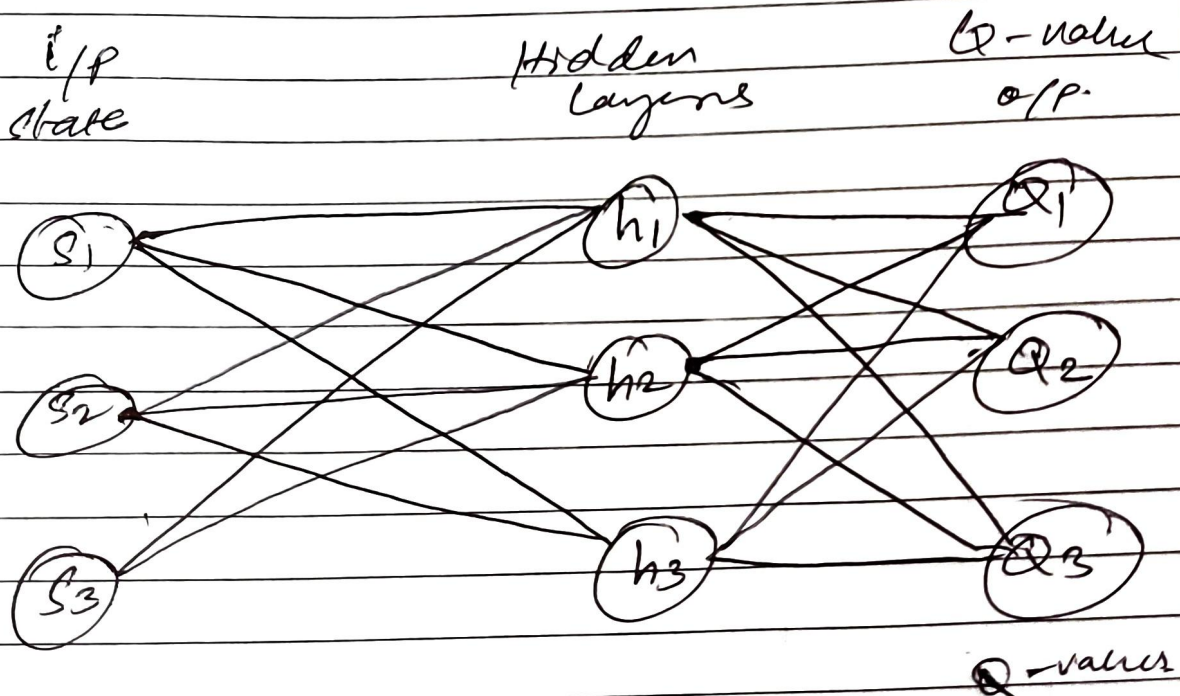
(Whereas Q-learning is off-policy)

Q value for the current state-action pair.

$\alpha \rightarrow$ learning rate

⇒ Deep Q Network:

- * Deep learning + Q Network.
- * use Neural Network.



- * Can solve complex problems
- * Real world complex problems.

Q) Lambda(λ) in RL :

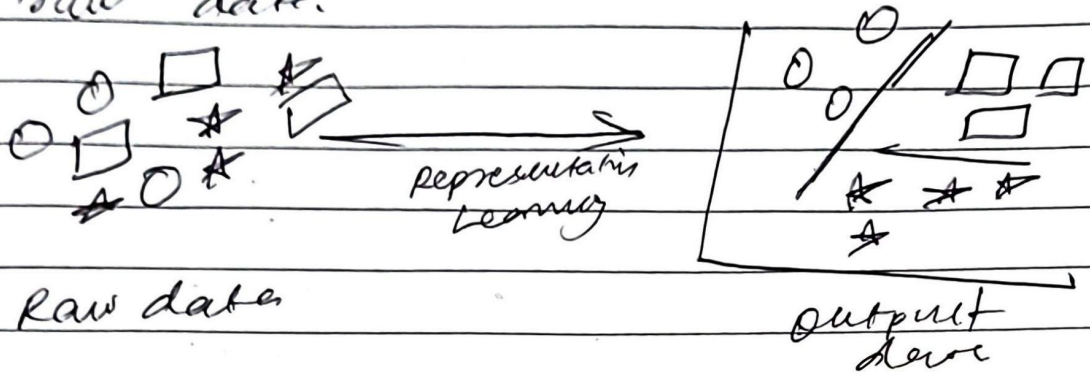
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- To get faster reward.
- Provides smooth Balance.
- We need to assign credit to earlier decisions.

module - 4

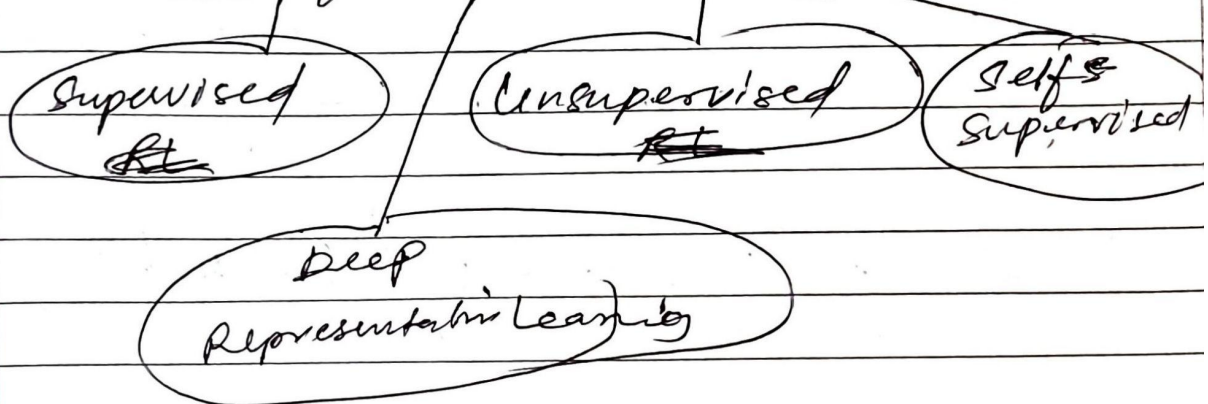
→ Representation Learning:

Branch of ML that focuses on automatically discovering meaningful features (representations) from raw data.



* It enable DL architectures

→ Types of Representation Learning:



used in Classification, Regression, NLP and etc.

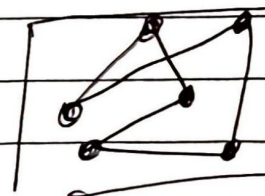
→ Graph-Based Learning:

mathematically, defined as:

$$G = (V, E)$$

$V \rightarrow$ set of nodes (vertices)

$E \rightarrow$ set of edges (connections)



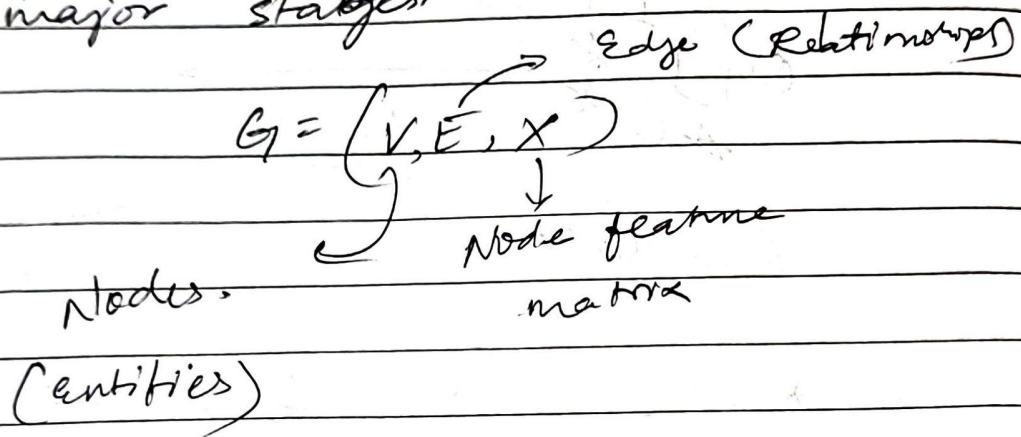
* Data points are ~~not~~ ^{inter}dependent

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* Connectivity influences prediction.

→ ML on Graphs:

① Graph construction in 5 major stages.



Eg: Social Network → Users (Nodes),
friendship (Edges)

Traffic Network → Intersections (Nodes),
Roads (Edges)

→ Kernel methods / Graph Kernel:

* ML techniques that measure similarity b/w data objects.

* Used to compare graphs, nodes, and etc.

* Tells us how similar two graphs are!

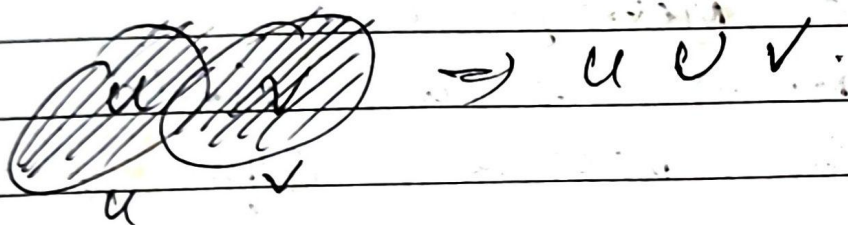
Use cases: Social Network analysis like Instagram, Facebook, etc.

neighbourhood overlap detection

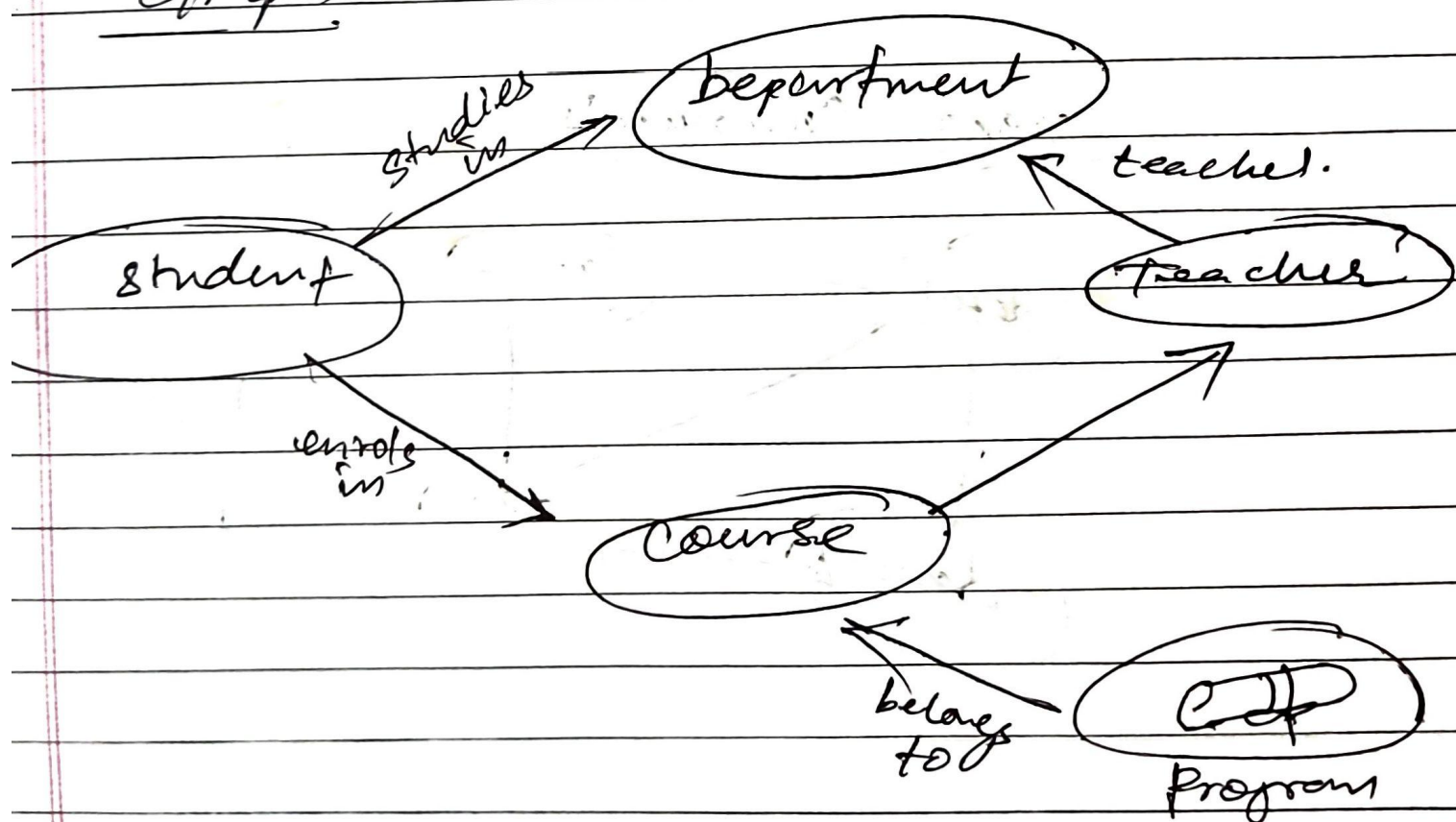
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$$NO(u, v) = \frac{|N(u) \cap N(v)|}{|N(u) \cup N(v)|}$$



→ multirelational data & Knowledge Graphs



Module - 5

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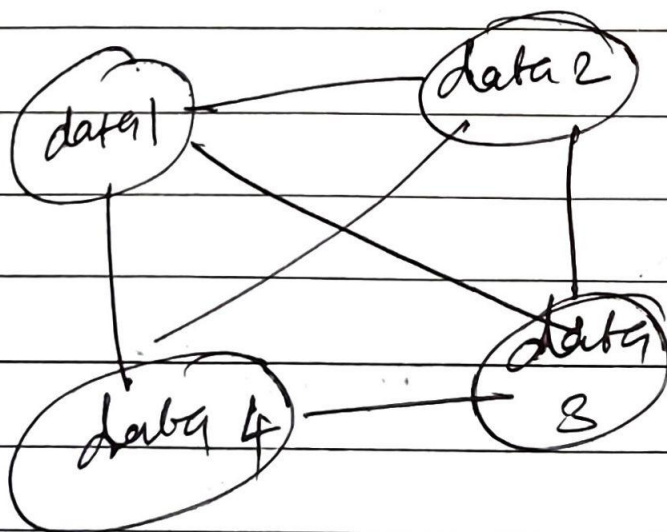
→ Graph Neural Networks :

* It's a DL model designed for graph-structured data, where data items are connected by relationships.

→ Neural Message Passing :

means each node in a graph receives info from its nodes, combines it with own info & then updates its representation.

→ Graph Convolution :



Refer queue also